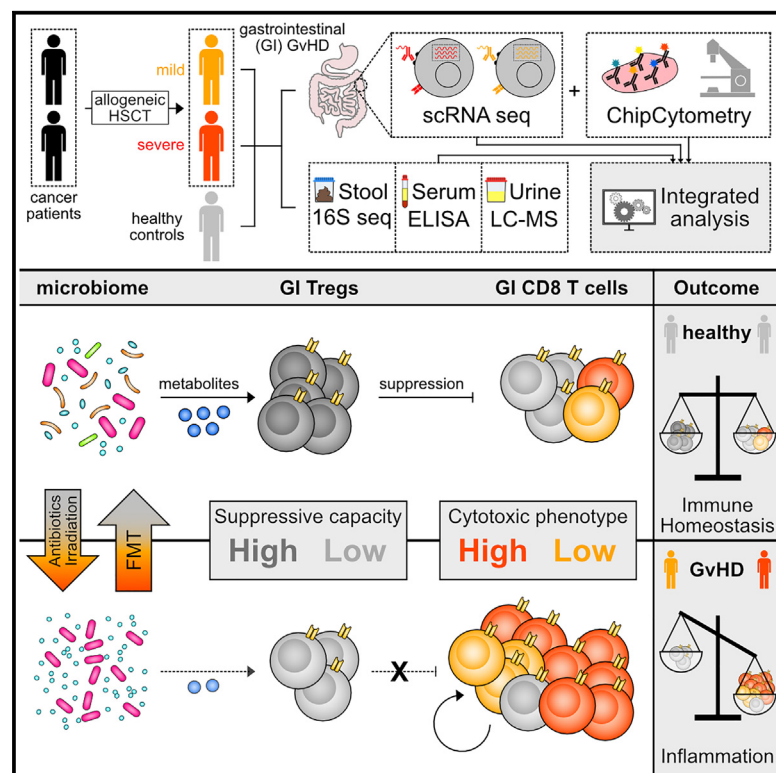


Multimodal immune cell phenotyping in GI biopsies reveals microbiome-related T cell modulations in human GvHD

Graphical abstract



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In brief

Jarosch et al. describe a link between clonally expanded CD8 T cells and GvHD severity, as well as the potential of GvHD suppression via microbiome-dependent Tregs. These findings support the concept of microbiome interventions as a GvHD treatment option and can serve as a starting point for the development of novel therapies.

Highlights

- Antibiotic exposure correlates with reduced Treg abundance in the gut
- Peripheral gut Treg differentiation is altered in antibiotic-exposed individuals
- GvHD severity associates with clonal expansion of activated CD8 T cells
- GvHD-associated clones are persistent over time and distributed within the GI tract



Article

Multimodal immune cell phenotyping in GI biopsies reveals microbiome-related T cell modulations in human GvHD

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SUMMARY

Acute graft-versus-host disease (aGvHD) is a significant complication after allogeneic hematopoietic stem cell transplantation (aHSCT), but major factors determining disease severity are not well defined yet. By combining multiplexed tissue imaging and single-cell RNA sequencing on gastrointestinal biopsies from aHSCT-treated individuals with fecal microbiome analysis, we link high microbiome diversity and the abundance of short-chain fatty acid-producing bacteria to the sustenance of suppressive regulatory T cells (Tregs). Furthermore, aGvHD severity strongly associates with the clonal expansion of mainly CD8 T cells, which we find distributed over anatomically distant regions of the gut, persistent over time, and inversely correlated with the presence of suppressive Tregs. Overall, our study highlights the pathophysiological importance of expanded CD8 T cell clones in the progression of aGvHD toward more severe clinical manifestations and strongly supports the further development of microbiome interventions as GvHD treatment via repopulation of the gut Treg niche to suppress inflammation.

INTRODUCTION

Allogeneic hematopoietic stem cell transplantation (aHSCT) is a powerful treatment option for several human diseases, such as anemias, immune deficiencies, and hematological malignancies.¹ It allows us to replace the hematopoietic system of a diseased individual with one from a healthy donor and is the standard-of-care treatment for high-risk acute leukemias. The associated graft-versus-leukemia (GvL) effect leads to higher long-term survival and lower relapse rates compared with classical cancer therapies; however, it frequently comes with the risk of developing graft-versus-host disease (GvHD), a potentially life-threatening immune reaction against the recipient's organs that eventually limits treatment success.¹

Gastrointestinal (GI) GvHD is initiated by the activation of antigen-presenting cells by danger signals (danger-associated

molecular patterns, DAMPs) and microbiota signals (pathogen-associated molecular patterns, PAMPs) as a result of barrier damage and bacteria translocation induced by aHSCT pre-conditioning, which is followed by the activation of donor T cells that finally mediate tissue injury.^{2,3} This tissue damage represents one of the hallmarks of GvHD, and it is detectable histologically via apoptosis of epithelial cells⁴ and serologically by elevated circulating levels of Reg3a.⁵ The GvHD incidence (35%–50%)⁶ after aHSCT increases with the number of mismatched human leukocyte antigen (HLA) alleles,⁷ and minor histocompatibility antigens⁸ between the donor and the recipient have been linked to disease severity, indicating that alloreactive mechanisms are at the basis of disease development.

Both CD4 and CD8 T cell lineages are involved in aGvHD progression and most likely equally needed to mount aGvHD in humans.⁹ While T cells reconstitute early within the first few months



after transplantation,¹⁰ the T cell receptor (TCR) repertoire remains severely skewed for at least the first year post-transplant.^{11,12} Furthermore, the emergence of expanded T cell clones and the consequent lowering of TCR repertoire diversity have been associated with the incidence of aGvHD.^{13–15} This knowledge opened the avenue to using the monitoring of TCR repertoire dynamics for early aGvHD diagnosis, but the importance of these expanded T cell clones in the etiology of GvHD remains unclear. On the one hand, the actual clonality and identity of the GvHD-related TCR repertoire is still undefined, as TCR diversity has been determined only according to unique TCR β -chain sequences rather than fully α/β -paired TCRs. On the other hand, knowledge of the spatiotemporal distribution of these clones is very limited as analyses of tissue-infiltrating, rather than circulating, T cells have been rarely performed so far in human settings.

Alloreactive donor T cells are a driving factor in GvHD development, but there are certainly additional mechanisms involved in setting the threshold for T cell activation as well as disease progression, as suggested by the anatomical restriction of GvHD manifestations and the strong dependency of alloreactivity on mechanisms associated with tissue inflammation and tolerance.^{16,17} The three main GvHD-affected organs are the liver, the skin, and the GI tract,² all of which are exposed to bacteria and/or microbial components and metabolites. It was shown in early days that the absence of the gut microbiome (in either germ-free or antibiotic-decontaminated animals) can partially rescue mice from GI GvHD.^{18,19} However, when this setting was mimicked in humans by antibiotic-mediated gut decontamination and isolation of individuals, no clear benefit was observed,²⁰ indicating a more complex and species-specific interaction between the host and the microbiome. This situation is further complicated by the use of broad-spectrum antibiotics to treat septic events in immunocompromised individuals. Recent studies, beyond confirming the association between GvHD risk and a decreased global microbiome diversity,^{21–23} also identified a reduction or increased abundance of bacterial species^{21,24–26} as specific factors contributing to aGvHD. Short-term detrimental effects of bacteria are often linked to translocation²⁷ or outgrowth of pathogenic strains, whereas long-term harmful effects are marked by dysbiosis and the loss of protective metabolites. Microbial metabolites, like short-chain fatty acids (SCFAs) and especially butyrate and propionate, have been correlated with a reduced GI GvHD risk and severity.²⁸ Potential mechanisms for the protective effects of SCFAs are the support of epithelial barrier integrity²⁹ and regulatory T cells (Tregs).^{28,30,31} Indeed, in addition to maintaining the gut Treg niche, SCFAs can also induce the peripheral differentiation of conventional CD4 T cells in so-called peripherally induced Tregs (pTregs).^{30,31} pTregs are themselves essential for mucosal tolerance,³² and low levels have been associated with GvHD development.^{33,34}

Based on this evidence, fecal microbiota transplantation (FMT), as a microbiome intervention strategy, is considered an attractive therapeutic option for the treatment of GvHD. However, although there have been promising first clinical observations, additional knowledge of the mechanistic connection between the microbiome and immune cell infiltrates in the setting of aHSC is required to improve and routinely apply microbiome

modulation as a safe treatment option for GvHD. The (often complex) phenotype of immune cells infiltrating GvHD-affected tissues has been explored only minimally in humans, mainly due to limited accessibility to disease-relevant biosamples and technical limitations on the multiplexing capacity of state-of-the-art immunohistochemistry.

To overcome the above-mentioned restrictions in the study of TCR repertoires and *in situ* immune cell phenotypes during aGvHD, in this study, we analyzed an exclusive cohort of biopsies after aHSC through two complementary technologies for single-cell analyses. First, we combined single-cell RNA sequencing (scRNA-seq) with the use of DNA-barcoded antibodies for in-depth, multiomics immune cell characterization. The dataset was further complemented by *in situ* phenotyping of immune infiltrates in GI tissues via ChipCytometry, a multiparameter imaging method that we previously implemented for formalin-fixed paraffin-embedded (FFPE) tissues.^{35,36} The cyclic immunofluorescence approach allows robust and high-quality imaging as well as automated signal quantification for multiple markers on the same tissue section. This combination of technologies allowed us to uncover phenotypes of conventional and regulatory T cells in combination with the TCR repertoire and spatial organization of infiltrates in human aGvHD samples and eventually to integrate this information with clinical information and 16S microbiome data.

RESULTS

Multiplexed scRNA-seq and tissue imaging for deep immune phenotyping of gastrointestinal biopsies from human aHSC-treated individuals

To gain insight into the mechanisms of GvHD pathophysiology, we analyzed a cohort of individuals who received HLA-matched or minor mismatched aHSC and were followed extensively after transplantation and over the course of GvHD. GI biopsies were taken at different time points during GvHD development (within 7 days of the first symptoms [onset] and more than 7 days after onset [course]) and from individuals who did not show any clinical GvHD sign (screening). We characterized the immune infiltration of those biopsies in depth, utilizing a multimodal analysis of scRNA-seq (from fresh biopsies) and ChipCytometry (on FFPE biopsies). Clinical data regarding GvHD biomarkers (Reg3a and ST2), antibiotic exposure, 16S microbiome data, and disease state were collected at the time of the biopsy (Table S1). GI GvHD severity was scored clinically according to the Glucksberg grading system³⁷ and grouped into mild (stages 0 and 1) and severe (stages 2–4) cases. As confirmation of the clinical scoring, higher levels of MAGIC biomarkers like ST2³⁸ and Reg3a⁵ and the MAP score were found in severe GvHD cases (Figure S1A).

To analyze GI immune cell infiltrates via scRNA-seq, we homogenized tissue biopsies and sorted CD45-positive leukocytes (Figure 1A). We were able to recover single-cell RNA transcriptomes and VDJ profiles from 31 biopsies collected from 26 individuals. Among the 31 biopsies, 19 were additionally stained with a cocktail of 130 DNA barcode-labeled antibodies, thus achieving multilevel single-cell data encompassing transcriptomes and surface protein expression. As a control cohort, we used healthy adjacent tissue from colon resections of colorectal cancer patients and integrated a published dataset of healthy

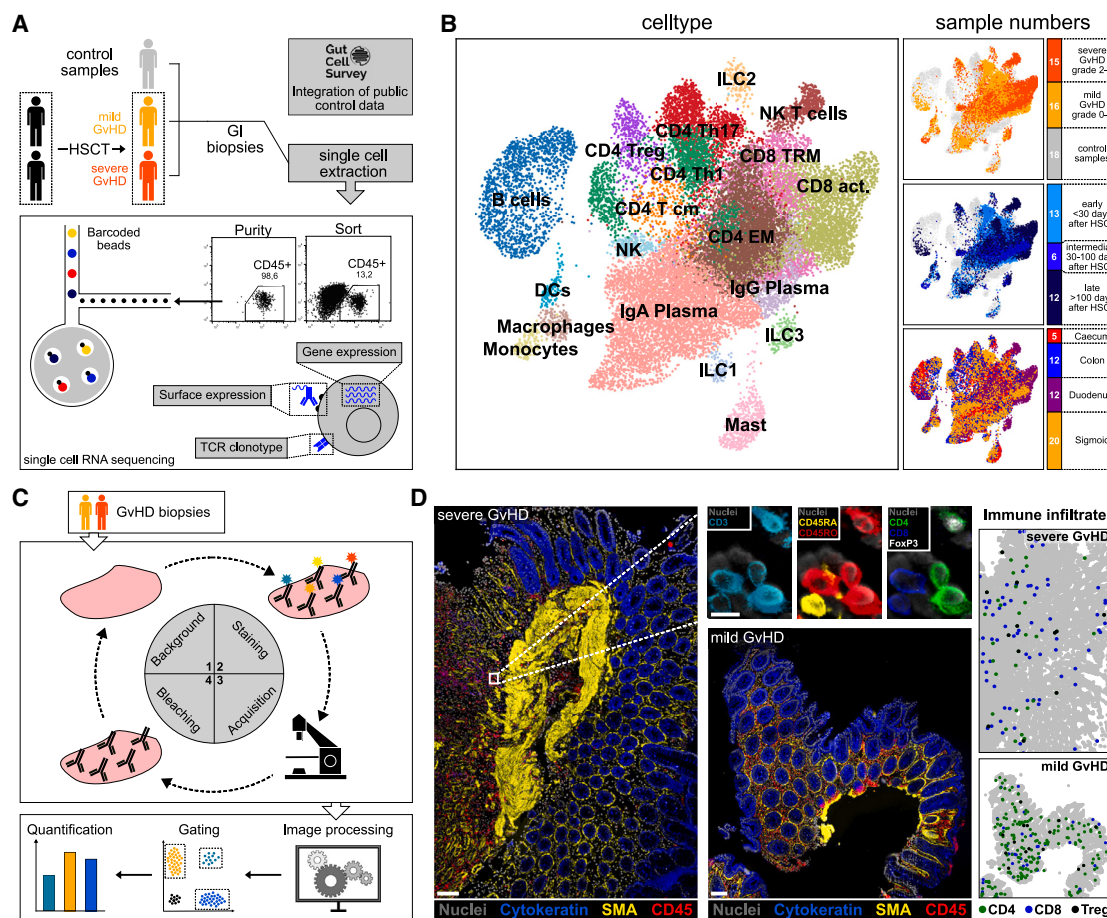


Figure 1. Multiplexed scRNA-seq and tissue imaging from gastrointestinal biopsies of human aHSCT-treated individuals

(A) CD45-positive immune cells were sorted from single-cell suspensions of fresh GI tissue samples and processed for scRNA-seq to analyze transcriptomic profiles, TCR repertoires, and surface marker expression.

(B) UMAP depiction of cell types (left) and distribution of samples according to their clinical characteristics (right).

(C) Schematic depiction of the processing of FFPE GvHD biopsies with the cyclic immunofluorescence-based ChipCytometry approach.

(D) Exemplary images of a severe (grade 3-4, sigmoid colon, left) and mild (grade 1, colon, middle) GvHD sample. Plots on the right show the distribution of CD4, CD8, and Treg cells within the tissue samples. Scale bars indicate 100 μ m in the overviews and 10 μ m in the zoom-in. SMA, smooth muscle actin. See also Figure S1 and Tables S1-S3.

individuals from the human gut cell atlas³⁹ (Table S2). Neighborhood embedding and clustering of the 22,008 cells (on average 450 cells per punch biopsy, see Table S3 for details on cell numbers) across all biopsies revealed different types of adaptive and innate immune cells (Figure 1B). The cell-type annotation was performed on conventional lineage marker genes and was cross-validated by DNA-tagged surface antibodies, which were part of the scRNA-seq readout. Further cell-subtype discrimination was achieved by phenotypic and activation markers (Figures S1B-S1D). We observed differential distribution of cells on the UMAP (Uniform Manifold Approximation and Projection), with enrichment of activated CD8 T cells in severe GvHD cases and at later time points. Interestingly, this cluster contained cells from different anatomic locations and several individuals, suggesting that the observed phenotype of this cluster could be attributed to T cell-intrinsic features rather than being dependent on external stimuli (Figures 1B and S1E).

The information on transcriptomic profiles was combined with the analysis of immune infiltrates within their spatial environment in GvHD biopsies via ChipCytometry. By iterative staining-acquisition-bleaching, we multiplexed up to 12 markers for this study, and the high image quality allowed us to perform automated generation of intensity values for the quantification of cell populations within the tissue (Figure 1C). The tissue architecture of the biopsies could be visualized by markers like Hoechst (nuclei), PAN-cytokeratin, and smooth muscle actin (SMA). T cell infiltrates were stained by specific markers for each subtype, allowing the analysis of T helper cells (CD3⁺CD4⁺), cytotoxic T cells (CD3⁺CD8⁺), and Tregs (CD3⁺CD4⁺FoxP3⁺), with further phenotypic characterization into resting and activated Tregs according to the expression of CD45RA and CD45RO,⁴⁰ respectively (Figure 1D). The visualization of CD4 and CD8 T cells already showed a difference in the CD4/CD8 ratio between the mild and the severe GvHD samples, with an increased CD8 T cell infiltration in

the latter (Figure 1D), consistent with what we observed in single-cell transcriptomic data (Figure 1B).

Altogether, we generated two datasets from GI biopsies of aHSC-treated individuals to facilitate in-depth analyses of infiltrating cell populations with single-cell, as well as spatial, resolution.

GvHD severity and antibiotic exposure correlate with increased CD8 T cell infiltration and reduced Treg abundance in the gut

To assess differences in immune cell composition between samples, we investigated the cell-type frequency distributions in the individual biopsies from the scRNA-seq dataset. We observed increased frequencies of the T cell compartment, both CD4 and CD8 T cells, and reduced levels of the B cell compartment (B cells, plasma cells) in aHSC-treated individuals compared with the healthy control cohort (Figure 2A), the latter likely related to a slow B cell reconstitution after transplantation⁴¹ and GvHD-related damage to the B cell niche.⁴² Interestingly, the enrichment of CD8 activated T cells (CD8 act.) was clearly visible in almost every individual with severe GvHD in comparison with the two other groups as well as across distinct anatomical sites of the GI tract (Figure 2A). Quantification of these differences confirmed the higher frequencies of B cells, plasma cells, and mast cells in control samples, whereas monocytes, CD4 memory cells, and CD8 activated T cells were more elevated in aHSC samples. Comparing mild and severe GvHD, we found activated CD8 T cells and regulatory CD4 T cells to be associated with GvHD severity (Figures 2B, 2C, and S2A), with no significant differences detectable between biopsies taken at disease onset or during progression (Figure S2B). When we looked at the interplay between CD8 activated T cells and Tregs, we observed a strong correlation in both healthy control and aHSC cohorts, which would indicate a balance between activation and immunosuppression within the gut environment (Figure 2D). However, the slope of this correlation was notably reduced in aHSC-treated individuals, suggesting an insufficient regulation of CD8-mediated inflammation by Tregs that might increase the risk of GvHD development.

To corroborate these findings *in situ* in a larger cohort, we made use of the ChipCytometry dataset. In total, we analyzed 52 GvHD biopsies from 40 aHSC-treated individuals and gated on marker intensities to define cell populations of interest, with a specific focus on T cells (Figure S2C). The quantification of cell densities showed significantly higher leukocyte and T cell infiltration in severe GvHD, mainly driven by increased CD8 T cell densities. CD4 infiltration, instead, did not differ according to GvHD severity, thus resulting in a skewed CD4/CD8 ratio in severe GvHD (Figure 2E). This finding was consistent with the trend of a reduced CD4/CD8 ratio observed within the scRNA-seq data (Figure 2C). Finally, although Treg infiltration density was not significantly higher in severe GvHD, it again correlated with the abundance of CD8 T cells (Figure 2F), as observed before. Altogether, these differences suggest that CD8 T cells might play a role in disease progression; indeed, their enrichment in frequency and infiltration density is associated with GvHD severity, where they numerically overwhelmed Treg infiltrating cells.

Like in every human study, we also had a complex set of clinical parameters to consider when interpreting our data. Thus, we

wanted to explore in an unbiased manner which variable(s) would explain the highest variance in our ChipCytometry dataset, since samples did not cluster according to GvHD severity on the principal component analysis (PCA) plot (Figure 2G). The principal components (PCs) of the data were associated with parameters of total CD4/CD8 T cell infiltration (PC1, 53% variance) and Treg infiltration or phenotype (PC2, 20% variance) (Figure 2H). To predict the factors contributing to the variance within the dataset, we calculated the fold change on both PCs for clinical parameters available for most of the individuals (Figure 2I and Table S4, see STAR Methods for calculation details). We found the time in terms of days after aHSC significantly connected to PC1 and thereby to T cell infiltration, indicating that the time of immune reconstitution should be considered when analyzing biopsies from different time points. In contrast, antibiotic exposure (ABX) and indoxyl sulfate levels were oppositely correlated with PC2 and Treg infiltration (Figures 2H, 2I, and S2D). This finding highlighted a possible influence of the microbiome on gut Tregs, as ABX and indoxyl sulfate were associated with low and high microbial diversity,²² respectively. ABX was defined as systemic treatment with broad-spectrum antibiotics at least 24 h ahead of biopsy sampling, while individuals from the non-ABX group did not receive broad-spectrum antibiotics at the time of biopsy or discontinued a minimum of 14 days before the biopsy (see STAR Methods for additional details). In our overall cohort, we could confirm that ABX is strongly linked to reduced microbiome diversity (Figure S2E). To finally understand the specific effect of ABX-induced microbiome alterations on Tregs, we investigated Treg tissue infiltration according to GvHD severity and ABX. Unexposed individuals showed a trend toward higher Treg infiltration in severe GvHD cases, to counterbalance inflammation, in sharp contrast to individuals receiving systemic antibiotic therapy (Figure 2J). Intriguingly, we additionally observed a significant drop in the frequency of Tregs with an activated phenotype (CD45RO⁺) in ABX individuals (Figure 2K).

Altogether, these results demonstrate a strong association between CD8 T cell infiltration, especially activated CD8 T cells, and GvHD severity. The increase in inflammation could be balanced by the concomitant induction of the Treg compartment. However, the diminished abundance of activated gut-infiltrating Tregs, presumably dependent on exposure to antibiotics and consequent alteration of the microbiome composition, might further unleash overshooting immune responses.

Regulatory T cell differentiation after aHSC is altered in antibiotic-exposed individuals

Previous analyses highlighted a connection of Tregs to the microbiome. To gain more insights, we subdivided the dataset according to FOXP3 expression and thereby extracted 207 FOXP3⁺ T cells from the aHSC cohort (26/31 biopsies, 21/26 individuals, Table S3) for in-depth analysis. Leiden clustering separated Tregs into three distinct clusters that were equally populated with cells from mild and severe GvHD cases (Figures 3A and S3A). Instead, Tregs from antibiotic-exposed individuals (referred to as “ABX-Tregs” from now on) were enriched in clusters 1 and 2, which were further associated with a reduced expression of FOXP3 (Figures 3A, S3A, and S3B). Analyses of differential gene expression and relative surface protein

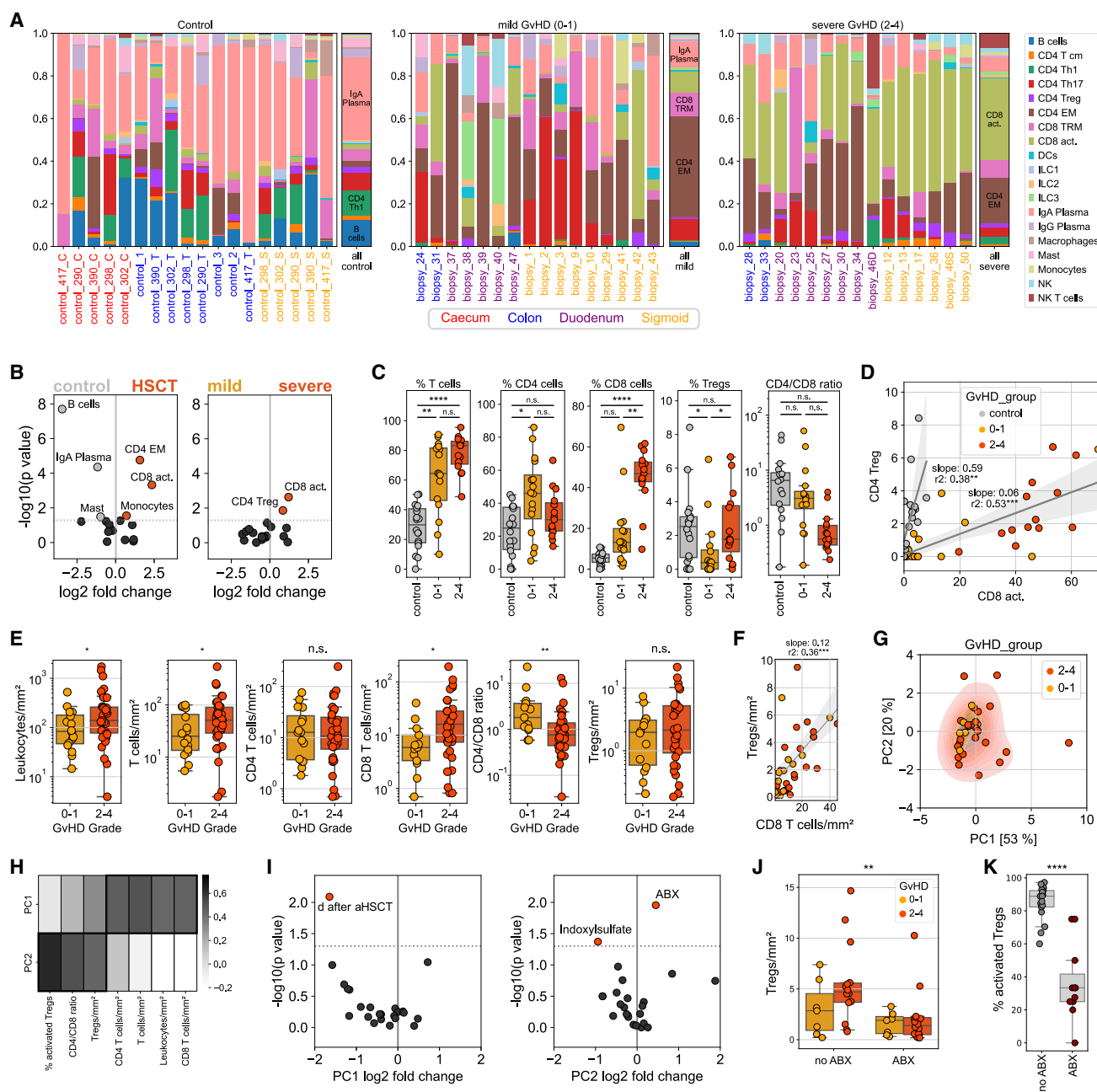


Figure 2. GvHD severity and antibiotic exposure correlate with increased CD8 T cell infiltration and reduced Treg abundance in the gut
(A) Cell-type frequencies from the scRNA-seq dataset for each individual sample. Axis labels are color coded according to the biopsy origin. Annotations in the bar graph refer to cell population with an abundance higher than 10%.
(B) Volcano plot showing significant changes in cell-type frequency. Each dot represents one cell type.
(C) Frequencies of gut-infiltrating T cell subsets. Each dot represents an individual biopsy.
(D) Linear regression analysis in aHsCT-treated individuals versus control samples.
(E–K) Automatically quantified ChipCytometry data. (E) Infiltration density of T cell subsets. (F) Linear regression analysis. Outliers (n = 8) exceeding 1.5 × IQR were removed. (G) PCA. The color indicates GvHD severity. (H) PCA loadings depicted as heatmap. (I) Volcano plots showing the fold change of clinical parameters along PC1 and PC2. Each dot represents one clinical parameter. (J) Treg frequencies. Statistical testing refers to antibiotic exposure alone. (K) Frequency of activated Tregs (CD45RO+). Statistical testing was conducted in (B) and (I) by Kruskal-Wallis H test, in (C) by one-way ANOVA followed by Dunn's multiple comparison with Holm-Bonferroni p value adjustment, and in (E), (J), and (K) by Mann-Whitney U test. Correlation analysis in (D) and (F) was done by Pearson's correlation (*p < 0.05; **p < 0.01; ***p < 0.001; ****p < 0.0001). See also Figure S2 and Table S4.

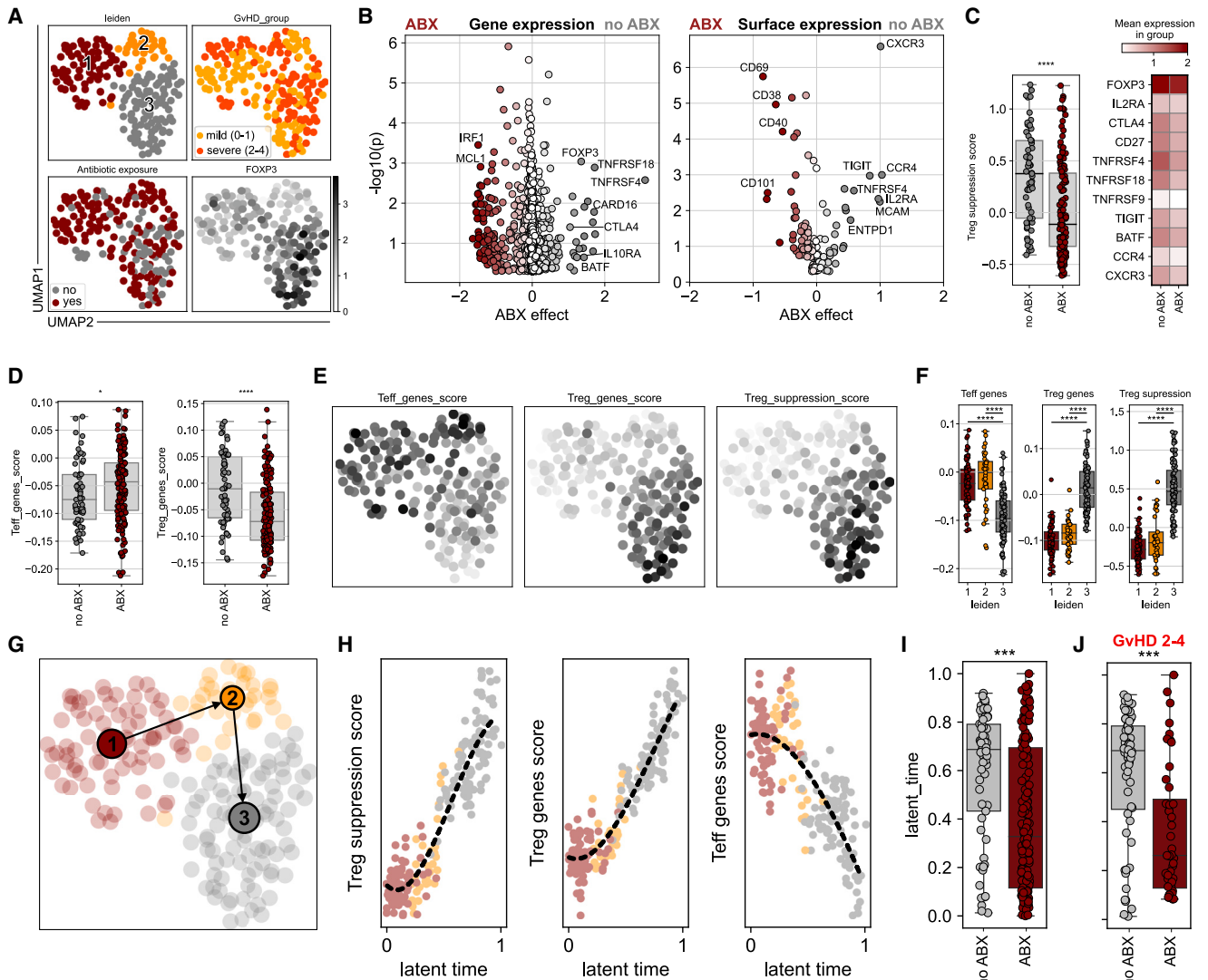


Figure 3. Regulatory T cell differentiation after aHSCT is altered in antibiotic-exposed individuals

(A) Leiden clustering of FOXP3⁺ cells extracted from the scRNA-seq dataset (26 biopsies).
 (B) Differential gene expression analysis (left) and surface expression analysis (right) according to antibiotic exposure (ABX).
 (C) Literature-defined genes were scored as "Treg suppression signature." Boxplots show the score according to ABX (left). Heatmap indicates the average expression for all individual genes of the score (right).
 (D) Teff and Treg gene scores are depicted according to ABX.
 (E) UMAPs depicting the distribution of the individual scores shown in (C) and (D).
 (F) Score quantification per Leiden cluster.
 (G) Differentiation path derived from RNA velocity analysis.
 (H) Latent time plotted against gene scores. The dashed line represents a polyfit of order 3.
 (I and J) Latent time plotted according to ABX for the whole dataset (I) or for severe GvHD cases (J). Each dot represents an individual FOXP3⁺ cell for all subplots except for (B), where each dot represents one gene. Differential gene expression analysis in (B) was performed using Welch's t test in diffpy. Statistical testing was conducted in (C), (D), (I), and (J) by Kruskal-Wallis H test and in (F) by one-way ANOVA followed by Dunn's multiple comparison with Holm-Bonferroni p value adjustment (*p < 0.05; ***p < 0.001; ****p < 0.0001). See also [Figure S3](#) and [Table S5](#).

abundance according to systemic ABX revealed an upregulation in genes and proteins associated with Treg signatures and functions in Tregs from untreated individuals (referred to as "no ABX-Tregs" from now on) (e.g., *FOXP3*, *TNFSF4*, *CTLA4* gene expression and CD134, CD25, CD39, CCR4, and CXCR3 surface expression) ([Figure 3B](#)). Gene ontology analysis of these en-

riched genes similarly revealed pathways of Treg functions and differentiation ([Figure S3C](#)). Next, we scored genes associated with Treg functionality based on literature research and found a significantly lower "Treg suppression" score in ABX-Tregs ([Figures 3C](#) and [S3D](#)), independent of the time point after aHSCT ([Figure S3E](#)). Furthermore, these Tregs showed a significantly

lower score also compared with the control cohort (Figure S3F), overall indicating a change in Treg phenotype in ABX-treated individuals toward reduced functionality.

Interestingly, ABX-Tregs showed increased expression of genes associated with conventional T cells (Figure 3B). When we scored public gene sets of effector T cells (Teff) and Tregs⁴³ (named “Teff genes” and “Treg genes” from now on; scored genes can be found in Table S5), we found an increased Teff gene score and decreased Treg gene score in ABX-Tregs compared with both no ABX-Tregs and control Tregs (Figures 3D, S3G, and S3H). These gene scores overlapped with some of the genes differentially expressed in our dataset, for which protein expression data were also available from DNA-tagged antibodies (Figures S3I and S3J), highlighting some interesting proteins that could be exploited for future (*in situ*) Treg characterization.

The three Treg clusters significantly differed in the described scores: clusters 1 and 2 shared a Teff-like signature, whereas cluster 3 had a significantly higher score for gene signatures associated with Treg functions (Figures 3E and 3F). We next asked whether these presumably less suppressive Tregs (cluster 1 and 2) would result from a de-differentiation of mature Tregs into conventional CD4 T cells or if they derive from circulating naive CD4 T cells that could not ultimately peripherally differentiate into Tregs. Analysis of transcriptional dynamics by RNA velocity allowed us to gain insights into the developmental direction of Tregs in the dataset. The suggested path started from Leiden cluster 1 via cluster 2 to cluster 3 (Figures 3G and S3K), which was the cluster with the highest Treg suppression score. Furthermore, the Treg suppression score and Treg gene score increased with latent time as well as with the number of spliced mRNAs from Treg signature genes, whereas the Teff gene score showed the opposite trend (Figures 3H and S3L). Remarkably, this directionality has even been detected within a single clonotype (Figure S3M). ABX-Tregs had a less differentiated phenotype, as shown by a significantly lower latent time (Figure 3I), also within the severe GvHD-associated Treg population (Figure 3J).

Taken together, these findings indicate an antibiotic-dependent phenotypic signature of gut-infiltrating Tregs, where ABX seems to interfere negatively with peripheral differentiation of conventional T cells into suppressive Tregs regardless of the GvHD status.

Specific microbial taxa associate with suppressive Treg phenotypes

In light of the connection between ABX and microbiome alteration, and the associated effect on Treg phenotype, we next investigated the possible involvement of specific bacterial taxa in the reconstitution of a suppressive Treg niche. In our 16S rRNA sequencing data from biopsy-matched stool samples ($n = 52$, Table S1), we did not observe significant clustering in the beta diversity plots according to GvHD severity, ABX, or time point after aHSCT, which might be due to the low number of individuals for a human microbiome comparison (Figure S4A). Nevertheless, since the previous data suggested a lack of an appropriate microbial stimulus for peripheral Treg development after ABX, we checked whether specific bacterial taxa are linked

to the previously mentioned scores defining Treg phenotypes. Therefore, we grouped individuals according to the presence or absence of each specific taxon and calculated the fold change of the gene scores in Tregs from these two groups. K-means clustering regarding the correlation between bacteria taxa and the Treg gene score, the Teff gene score, and the Treg suppression score identified two bacteria clusters supportive of Treg development, whereas another two were connected to the Teff gene score (Figures 4A, 4B, and S4B). Taking a deeper look into the composition of these clusters, Clostridia and Bacteroidia seemed to be enriched in the Treg-supporting clusters, whereas bacteria from the Gammaproteobacteria class were found enriched in Teff-supporting clusters (Figure 4C). These findings are in line with existing evidence describing the positive effect of specific bacterial classes in the microbiome on gut Tregs.^{44,45}

Treg induction can be mediated by bacterial metabolites, especially SCFAs, including butyrate and propionate.^{30,46} We, therefore, checked for known metabolites produced by the identified taxa in the Virtual Metabolic Human database (VMH)^{47,48} and indeed found enrichment in butyrate and propionate-producing taxa in the Treg-supporting clusters (Figure 4D). LDA effect size analysis (LEfSe) led to the detection of differential species according to ABX treatment of the individual. Here we found some overlap between taxa enriched without ABX and Treg differentiation-supporting taxa (Figure S4C).

Altogether, these observations highlight the potential of the microbiome in Treg induction and support the idea of microbiome manipulation as clinical intervention for GvHD treatment.

Conventional T cells show enhanced clonal expansion in severe GvHD and longevity across different gastrointestinal anatomical locations

Regarding GvHD pathophysiology, we have seen that conventional T cells are associated explicitly with severe disease (Figure 2). With the aim of analyzing this compartment deeper, we subdivided the dataset according to Leiden clustering and analyzed T cell clusters containing 7,218 cells from our scRNA-seq cohort (all samples included). First, we were able to confirm the trend of CD8 enrichment in severe GvHD cases (Figure 5A). We next performed differential gene expression analysis according to GvHD severity and found a cytotoxic effector gene signature (*NKG7*, *GZMA*, *GZMK*, *HAVCR2*, *CST7*) associated with T cells from severe GvHD, presumably due to the intrinsic enrichment of activated CD8 T cells in those individuals. T cells from mild GvHD, in contrast, showed increased expression of memory-associated genes (*IL7R*, *KLRB1*), in line with the generally lower inflammation (Figure 5B). Further analyses within the CD8 and CD4 subsets revealed an enrichment in cytotoxic genes primarily for CD8 T cells, thus confirming the more cytotoxic effector phenotype of CD8 T cells in severe GvHD cases (Figure S5A).

In the gene set enrichment analysis (GSEA) of genes upregulated in T cells from severe GvHD cases, we found “graft-versus-host disease” as a top hit among other pathways associated with transplantation and autoimmunity (Figure 5C). When we scored the genes from this pathway in all individual CD4 or CD8 T cells for each biopsy, we found a higher GvHD score in CD8 than in CD4 T cells, in addition to the expected correlation

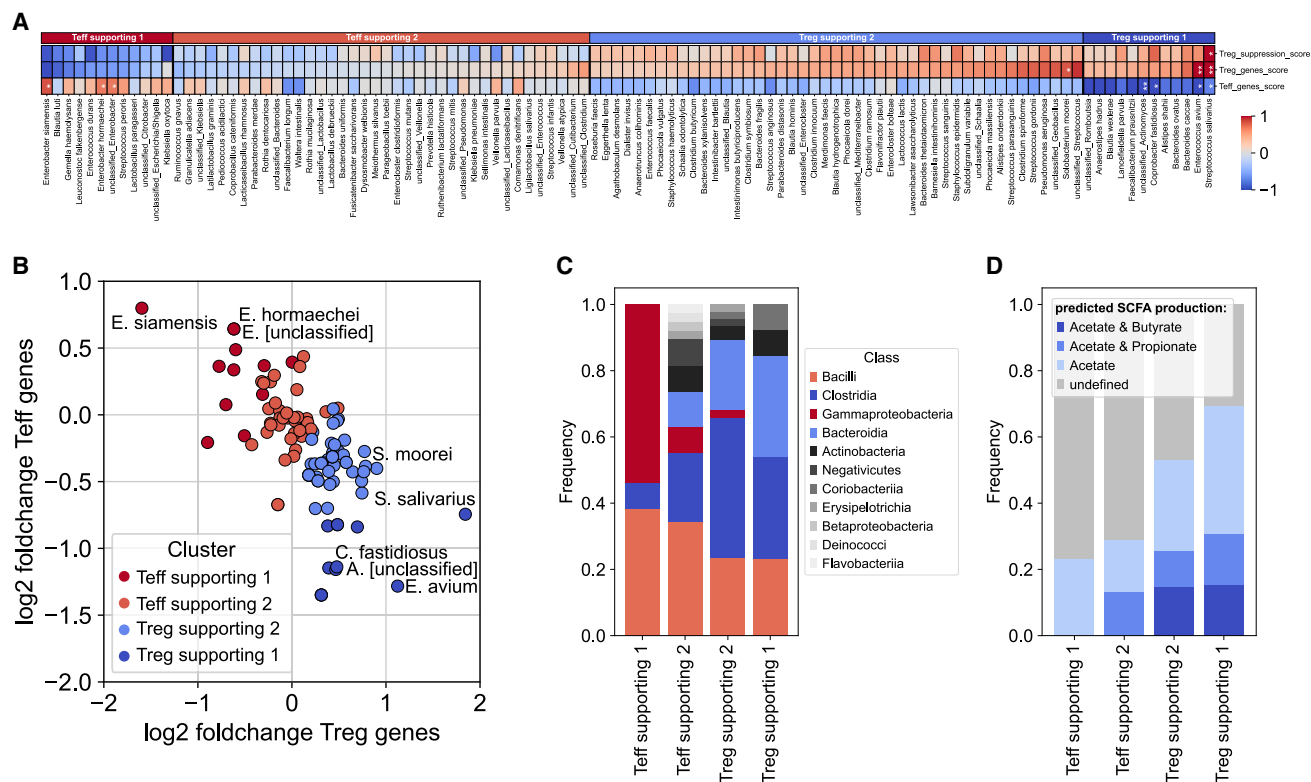


Figure 4. Specific microbial taxa associate with suppressive Treg phenotypes

(A) Presence of bacterial taxa was associated with Treg scores defined in Figure 3. Taxa were subdivided by K-means clustering according to their association pattern with gene scores.

(B) Bacterial taxa are plotted according to their effect on the Treg gene score and the Treg gene score. The color coding refers to the K-means clustering.

(C) Bacterial class abundance within the individual clusters.

(D) Ability of SCFA production by the individual bacterial species was retrieved from the Virtual Metabolic Human database, and the frequency of metabolite-producing species is depicted per cluster. Statistical testing in (A) was conducted by Kruskal-Wallis H test (* $p < 0.05$; ** $p < 0.01$). See also Figure S4.

with clinical GvHD severity of the respective individuals (Figure 5D). This is in line with our previous finding that primarily CD8 T cells are associated with severe GvHD (Figures 1E and 2D) in our cohort. CD8 T cells, and especially those with an activated phenotype, but not CD4 T cells were furthermore highly enriched in individuals with high plasma levels of Reg3a (Figure S5B). This association emphasizes the importance of CD8 T cells in GvHD pathophysiology since Reg3a is a clinical indicator for barrier disruption and a marker for GvHD severity.⁵

VDJ sequencing allowed us to investigate clonal expansion of the T cell compartment. In addition to a globally higher clonal expansion in aHSCT-treated individuals (Figure 5E), we found significantly increased clonality, and consequently decreased TCR diversity, also according to the degree of GvHD severity (Figure 5F) and independent of the time point after aHSCT (Figure S5C). Intriguingly, we found some highly expanded clonotypes with more than 15 cells per TCR (Figure S5D). The absolute clonal expansion was again driven mainly by the CD8 T cell compartment. In contrast, CD4 TCR repertoire diversity remained stable between mild and severe GvHD, except for two biopsies that showed a marked skewed repertoire due to a few highly expanded clonotypes within the conventional CD4 T cell subset. We did not observe clonal expansion for Treg cells, possibly

due to the overall low number of detected cells (Figures 5F, 5G, S5E, and S5F). The level of CD8 clonality supports antigen-specific expansion, and it has further been strengthened by the detection of several TCR clusters based on amino acid sequences, probably recognizing a similar epitope (Figures S5G and S5H).

Notably, our human biosample cohort contained biopsies collected either longitudinally or from different locations in the GI tract of three individuals. When we calculated the TCR repertoire overlap of these biopsies, we found some intra- but not inter-individual similarities (Figure 5H). For patient 56, we analyzed two biopsies taken on day 154 from the sigmoid (biopsy 17) and on day 350 from the duodenum (biopsy 27) (Figure 5H). Although deriving from anatomically distant sites and sampled over the course of almost 200 days, we found 21 clonotypes overlapping between the biopsies, all of them from the CD8 T cell compartment (Figures 5I and 5J). These obviously broadly relevant clonotypes shared, most of the time, a rather activated phenotype (Figures 5J and 5K). In addition to the detection of the same clonotypes between biopsies, the TCR repertoire of this individual showed a high proportion of TCRs sharing similar motifs as detected by amino acid alignment (Figure S6G). We could further corroborate this finding of GI distribution of expanded T cell clones in another pair of biopsies from sigmoid and duodenum

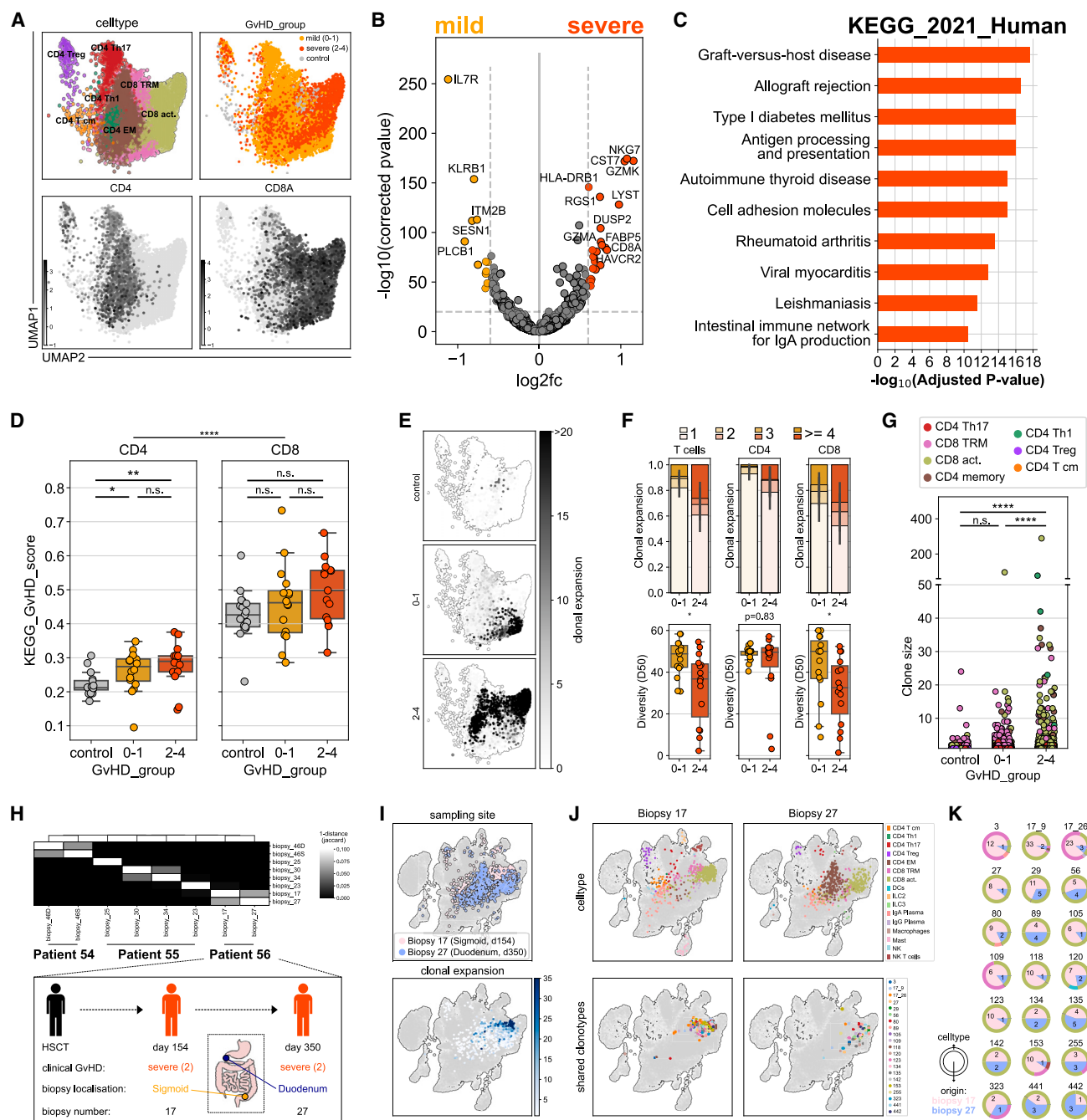


Figure 5. Conventional T cells show enhanced clonal expansion in severe GvHD cases and longevity across different gastrointestinal anatomical locations

(A) UMAPs of the T cells extracted from the total scRNA-seq dataset.
(B) Volcano plot showing differential gene expression analysis according to GvHD severity.
(C) Gene set enrichment analysis for the KEGG 2021 human database performed on genes more highly expressed in severe GvHD cases.
(D) Quantification of the GvHD score from (C). Each dot represents an individual biopsy.
(E) UMAPs showing the clonal expansion of T cells. The gray scale indicates the number of individual cells expressing the same TCR clonotype (same TRAC/ TRBC CDR3 region).
(F) Quantification of clonal expansion as frequency of T cells with a clone size of 1, 2, 3, and ≥ 4 . The different clone size groups are visualized by color shading (top). Quantification of TCR repertoire diversity is shown at the bottom.
(G) Absolute clone size for all individual clonotypes grouped by the GvHD severity.

(legend continued on next page)

taken at the same time point (patient 54, [Figure S6A](#)) and two of four duodenal biopsies taken at different time points over the course of severe GvHD (patient 55, [Figure S6B](#)).

Altogether, we observed that severe GvHD associates with, and is presumably driven by, highly activated and clonally expanded CD8 T cells, further characterized by a relatively long persistence and broad distribution across the GI tract.

Tregs might contribute to the suppression of CD8 T cell-mediated GvHD via spatial proximity

The two most striking observations of our study were the reduction in Treg numbers and the suppressive phenotype in individuals exposed to antibiotics and with microbiota disruption and the clonal expansion of CD8 T cells in severe GvHD. We finally analyzed whether and how the interplay between these two cell types might contribute to the progression of GvHD. For that, we quantified the T cell clone sizes within the group of severe GvHD cases according to the mean Treg suppression score of the corresponding individual. Intriguingly, we could observe that individuals showing a more pronounced suppressive phenotype of the Treg compartment had significantly less T cell clonal expansion for CD8 T cells but not CD4 T cells ([Figure 6A](#)). In light of the supportive role of the microbiome in Treg differentiation and functions ([Figures 3 and 4](#)), and consequently in mitigating GvHD development, we then investigated how therapeutic interventions targeting the gut bacteria composition could, in principle, act on the Treg/CD8 T cell balance. Thus, we analyzed biopsies from one severe GvHD-affected individual before and after FMT, which resulted in a significant increase in microbial diversity and complete response in GI GvHD even at 9 months of follow-up.⁴⁹ Using ChipCytometry, we characterized and quantified immune cell infiltrations within their spatial organization and detected a trend of decrease in CD8 T cell infiltration together with a higher Treg infiltration ([Figure 6B](#)). This higher Treg density led to a higher probability of interaction with CD8 T cells by spatial proximity ([Figure 6C](#)), as indicated by an overall significantly reduced distance of CD8 T cells to the nearest Treg after FMT treatment ([Figure 6D](#)).

These findings would suggest that Tregs might mediate general immunosuppression according to a balanced ratio with CD8 T cells, and their immunosuppressive potential can be tuned by microbiome composition. Further testing in larger cohorts of aHSCT-treated individuals receiving FMT are, however, necessary to confirm this putative mechanism.

DISCUSSION

This study successfully combined scRNA-seq and ChipCytometry imaging to generate a comprehensive clinical dataset for the char-

acterization of immune infiltration in an exclusive collection of GI biopsies from 60 aHSCT individuals. The 16S microbiome data from stool samples, clinical metadata, and tight clinical monitoring of GvHD biomarkers from urine and serum samples completed the multimodal dataset. Subsequent analyses showed a strong association between GvHD progression and CD8 T cell expansion, which might also be linked to the differentiation and suppressive potential of microbiome-inducible Tregs.

We observed higher T cell infiltration in the gut of GvHD-affected individuals, in line with existing studies associating levels of circulating T cells post-HSCT with the risk of GvHD.^{50,51} In our cohort, CD8 T cells rather than CD4 T cells correlated with GvHD severity, as indicated by the display of an activated phenotype, GvHD-related gene signatures, and increasingly higher density of tissue infiltration according to GvHD severity. In addition, CD8 T cell infiltration correlated with Reg3a levels, which is a marker for target organ damage.⁵ This increase in T cell density could be explained by the clonal expansion of a restricted number of TCR clonotypes, mainly from activated CD8 T cells, and consequent skewing of the TCR repertoire in particular for severe GvHD cases up to 350 days from aHSCT (the longest follow-up in our cohort). These data corroborated, in GI GvHD-relevant tissue sites—colon, duodenum, and sigmoid—previous studies that detected an abnormal TCR repertoire in the peripheral circulation of GvHD-affected individuals within at least the first year of transplantation.^{13,15–17} Furthermore, we found that the most abundant TCR clonotypes are broadly distributed within the GI tract, thereby complementing similar findings from pre-clinical mouse models^{52,53} and autopsies on severe GvHD.⁵⁴ The additional observation that these shared TCR clonotypes are also stable over time in longitudinal samples strengthened their relevance not only in the development of GI GvHD but also in its progression toward severe clinical manifestations.

Our evidence on TCR clonality, distribution, and longevity in GvHD-affected organs suggests that antigen specificity could play an important role in the setting of GI GvHD. Moreover, naive T cells seem to be more relevant for GvHD development than central memory or Teffs, indicating that antigen-specific priming is likewise important for disease development.⁵⁵ This is supported also by the highly divergent TCR repertoires among individuals and mice and in comparison with the pre-transplant donor cells,^{52,53} which could be explained by the clonal expansion of extremely low-frequency T cells from the naive pool. Different from previous studies that defined TCR repertoires only according to Vbeta diversity,^{13,15–17,52,53} we analyzed fully α/β -paired TCRs concomitant with their phenotypic signature, thereby shedding new light on the field of TCR repertoires in human GvHD with a multidimensional level of information. By

(H) Heatmap showing the repertoire overlap among all longitudinal samples (top). Schematic depiction explaining sample origins of biopsies from patient 56 (bottom).

(I) UMAPs depicting the distribution of the cells (top) and the clonal expansion (bottom).

(J) Cell-type clusters (top) and location of shared clonotypes (bottom).

(K) All 21 shared clonotypes are depicted with the number of cells from the individual biopsies (inner circle) and the assigned cell type (outer ring). Numbers above pie charts indicate the represented clonotype. Differential gene expression analysis in (B) was performed using Welch's t test in diffxpy. Statistical testing was conducted in (D) (CD4 versus CD8) and (F) by Kruskal-Wallis H test and in (D) and (G) by one-way ANOVA followed by Dunn's multiple comparison with Holm-Bonferroni p value adjustment (*p < 0.05; **p < 0.01; ****p < 0.0001). See also [Figures S5 and S6](#).

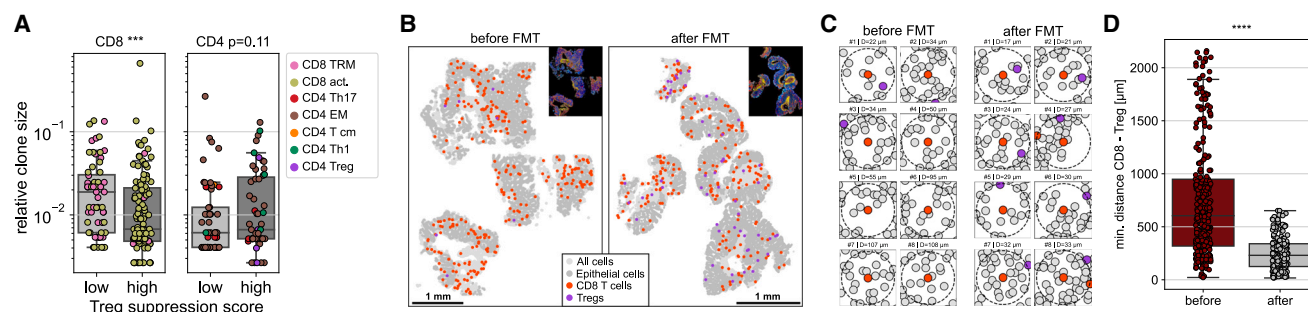


Figure 6. Tregs might contribute to the suppression of CD8 T cell-mediated GvHD via spatial proximity

(A) Relative clonal expansion (clone size divided by the number of T cells in the respective biopsy) of CD8 (left) and CD4 T cells (right) in severe GvHD cases grouped according to the individual's mean Treg suppression score.

(B) Computational spatial reconstruction of CD8 and Treg infiltration before and after FMT from ChipCytometry analyses.

(C) Top eight CD8 T cells with the closest distance to a Treg are shown before and after FMT treatment.

(D) The minimal distance between a CD8 T cell and the closest Treg is plotted before and after FMT for each CD8 T cell. Statistical testing in (A) and (D) was conducted by Kruskal-Wallis H test (***p < 0.001; ****p < 0.0001).

providing a clear identity of the putative GI GvHD-driving T cell clones, it would now be feasible to explore and identify the antigens driving this T cell expansion. In the case of alloreactivity, the CD8 T cell-dependent immune activation hints toward the recognition of minor histocompatibility antigens in our cohort, since most of the analyzed individuals received a fully HLA class I-matched transplant. In addition, other than allogeneic self-epitopes, the question remains if epitopes derived from bacteria, critical players in GI health and disease, could also be a factor.

In response to GvHD as an inflammatory event associated with clonal CD8 T cell expansion, we detected an increase also in Tregs, confirming the higher FoxP3 expression in GvHD tissue samples,⁵⁶ but with high precision in terms of cell subset definition. Indeed, FoxP3 alone does not unequivocally define a Treg, as it can be transiently expressed also by other cell types.⁵⁷ In addition, we could correlate the abundance of Treg and CD8 T cells as both frequencies and absolute infiltration densities, thus showing that the counterintuitive higher Treg infiltration in severe GvHD could be a direct response to T cell-mediated tissue inflammation. Nevertheless, we still observed an inadequate quantitative reconstitution of Tregs after aHSCt, as indicated by a lower correlation slope in these individuals. While the disproportionately high expansion of CD8 T cell clones is a clear cause of this altered balance between tolerance and inflammation, we also found that ABX associated with different infiltration and phenotypes of the gut Treg niche.

Antibiotics are well known to alter the gut microbiota, which in turns affects the Treg compartment. Indeed, microbiota metabolites strongly contribute to the overall maintenance of Treg homeostasis and to the generation of pTregs,^{28,30,31} which are the main mediators of gut tolerance. While these mechanistic connections are well accepted in pre-clinical models, there is a paucity of evidence in human settings.⁴⁶ In addition to limited availability of relevant biosamples, other reasons are the often-complex phenotype of human (p)Tregs and technological limitations in the simultaneous detection of multiple markers, in particular *in situ*. In our study, via the use of multiplexed tissue staining, we markedly improved the precision by which Tregs are defined

in the gut of aHSCt-treated individuals. We clearly showed a reduced Treg infiltration and frequency of activated CD45RO⁺ Tregs upon antibiotic administration, independent of the GvHD status. We could support this observation of reduced priming by scRNA-seq multiomics. First, we found that FoxP3⁺ T cells of individuals exposed to antibiotics show a less immunosuppressive phenotype together with a low level of FoxP3 expression. Second, the analysis of transcriptional dynamics within the FoxP3⁺ compartment revealed a differentiation trajectory moving from cells with gene signatures more typical of T effector cells to cells with a strong Treg phenotype, which would resemble the peripheral differentiation path of pTregs. Although we could not unequivocally identify pTregs in our dataset, the described findings altogether would suggest an antibiotic-mediated interruption of pTreg differentiation. Experimental validations are, however, necessary to validate this hypothesis.

As mentioned before, the microbiota, and in particular bacteria species producing butyrate and propionate, are highly Treg supportive. In our dataset, we found SCFA-producing taxa, especially from the Clostridia and Bacteroidia classes, associated with more differentiated and suppressive Treg phenotypes. Many of the associated species have been previously described to reduce inflammation and induce tolerance or regulatory mechanisms (e.g., Treg supporting cluster 1: *Streptococcus salivarius*,⁵⁸ *Faecalibacterium prausnitzii*,⁵⁹ *Blautia wexlerae*,⁶⁰ and *Bacteroides ovatus*⁶¹), confirming a potential role in GvHD suppression. Connected to the effect of ABX in inducing microbial dysbiosis, we here provide the first evidence that could overall support the mechanism by which ABX interferes with Treg differentiation potentially due to a reduced microbiome-dependent peripheral Treg induction also in humans.

Finally, we could link our findings about clonally expanded CD8 T cells with the observed association between ABX and Treg phenotype, as we found a reduced CD8 T cell expansion in individuals with a higher suppressive potential of Tregs. Notably, we could also observe in a single steroid-refractory GvHD case that complete response after FMT was accompanied by an increased Treg infiltration and consequently higher

likelihood of interactions between Tregs and CD8 T cells due to reduced spatial proximity. Although no conclusions can be drawn from a single case, these observations support the use of FMT for GvHD treatment. FMT as microbiome intervention has been shown to partially increase survival rates in the first clinical trials and might therefore represent a promising treatment option for GvHD.⁶² Indeed, FMT can be curative in more than 40%–50% of steroid-refractory GvHD, which otherwise have mortality rates of 70%–80%.^{63,64} Furthermore, FMT has even been proven safe as prophylactic treatment for GvHD.⁶⁵

In summary, this study gives essential insights into the mechanisms of GvHD pathophysiology and the connections between the microbiome, immune infiltrates, and clinical outcomes in aHSCt-treated individuals. The relevance of antigen-specific, clonally expanded CD8 T cells in GvHD progression highlights the importance of identifying antigens driving GvHD, which might represent a new clinical intervention point. On the side of Tregs, the data support microbiome interventions for GvHD treatment or even prophylaxis. Indeed, the increase in microbial diversity and abundance of SCFA-producing species could support the reconstitution of a functional gut Treg niche, itself contributing to the control of conventional T cell clonal expansion and GvHD severity.

Limitations of the study

The analyzed cohort is unique regarding tissue samples from human GvHD but is limited in size regarding the number of profiled cells per individual and the microbiome data. For this reason, the here-identified signatures need to be confirmed in larger cohorts, considering that improvements in the yield of cells extracted from gut biopsies are difficult to achieve. Also, cohorts with a more homogeneous profile of clinical parameters are advisable, to systematically pinpoint the impact of specific interventions—e.g., ABX, steroid treatment—on T cells, microbiota, and eventually GvHD. Similarly, a larger number of FMT-treated individuals need to be analyzed to confirm our preliminary model of interaction between the microbiome, Tregs, and clonally expanded conventional T cells. Finally, further efforts are required to achieve more precise *in situ* phenotyping of Treg populations, in particular for pTregs. The combination of mRNA *in situ* hybridization with surface marker staining³⁶ for the detection of scarce or difficult-to-detect markers (e.g., CTLA-4, IL-10) would already serve for this purpose.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.xcrm.2023.101125>.

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AUTHOR CONTRIBUTIONS

Conceptualization, D.H.B., E.D., and E.H.; data curation, S.J., J.K., S.G., A.H., and E.M.; formal analysis, S.J. and A.H.; funding acquisition, D.H.B., E.D., E.H., and E.M.; investigation, S.J., J.K., S.G., and M.H.; methodology, S.J., J.K., and A.H.; project administration, D.H.B., E.D., E.H., and E.M.; resources, E.T.O., D.G., M.E., K.-P.J., D.W., and E.M.; software, S.J.; supervision, A.G., D.H.B., E.D., E.H., M.E., and H.P.; visualization, S.J.; writing – original draft, S.J.; writing – review & editing, S.J., E.D., and E.H. All authors read, revised, and approved the final draft.

DECLARATION OF INTERESTS

D.H.B. is a co-founder of STAGE Cell Therapeutics GmbH (now Juno Therapeutics/Bristol Myers Squibb) and T Cell Factory B.V. (now Kite/Gilead). D.H.B. has a consulting contract with and receives sponsored research support from Juno Therapeutics/BMS. H.P. is a consultant for Gilead, AbbVie, Pfizer, Novartis, Neovii, Janssen-Cilag, Servier, and Bristol Myers Squibb. H.P. has received research funding from Bristol Myers Squibb. E.H. is a consultant of Maat Pharma, PharmaBiome, Novartis, Neovii, and Medac.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Anti-human Pan-Cytokeratin (AF488)	BioLegend	RRID: AB_2616664
Anti-human CD4 (AF488)	R&D systems	RRID: AB_2728839
Anti-human Vimentin (AF488)	BioLegend	RRID: AB_2650955
Anti-human Vinculin (AF488)	Thermo Fisher Scientific	RRID: AB_2574473
Anti-human Foxp3 (PE)	Thermo Fisher Scientific	RRID: AB_1944444
Anti-human Ki-67 (PE)	BD Biosciences	RRID: AB_2266296
Anti-human CD45RA (PE)	BioLegend	RRID: AB_314412
Anti-human CD8 (PE)	Santa Cruz	RRID: AB_1120718
Anti-human CD45RO (PE)	BioLegend	RRID: AB_314422
Anti-human CD3 (PE)	Beckman Coulter	Cat#: M0701
Anti-human SMA (eFluor570)	Thermo Fisher Scientific	RRID: AB_2573630
Anti-human CD45 (PerCP/Cy5.5)	BioLegend	RRID: AB_893338
Anti-human Ki-67 (PerCP/Cy5.5)	BD Bioscience	RRID: AB_10611574
Anti-human CD45RA (PerCP/Cy5.5)	BioLegend	RRID: AB_893357
Anti-human CD3	Thermo Fisher Scientific	RRID: AB_149924
Anti-human CD45 (Pacific Blue)	Dako	Cat#: IM2467
Anti-rabbit IgG (PE)	Biolegend	RRID: AB_2563484
Biological samples		
Healthy adjacent tissues from colon resections	Molecular Tumor Biology (Prof. Dr. Klaus-Peter Janssen) at the Dept. of Surgery (TUM)	N/A
Biopsies from aHSCt patients	University hospital of Regensburg (Prof. Ernst Holler)	N/A
Chemicals, peptides, and recombinant proteins		
Hoechst	Thermo Fisher Scientific	Cat#: H3570
Ethanol absolute, 1 % MEK	Carl Roth	Cat#: K928
Ethanol 70%, 1 % MEK	Carl Roth	Cat#: T913
Roticlear	Carl Roth	Cat#: A538
Tween20	Carl Roth	Cat#: 9127
Tris(hydroxymethyl)aminomethane (TRIS)	Carl Roth	Cat#: 9429
Ethylenediaminetetraacetic acid (EDTA)	Carl Roth	Cat#: X986
Sudan Black B	Sigma Aldrich	Cat#: 199664
True Black	Biotium	Cat#: 23007
Dulbecco's phosphate-buffered saline (DPBS)	PAN-Biotech	Cat#: P04-36050P
Sodium Borohydride	Sigma Aldrich	Cat#: 71320
Hydrogen peroxide (H ₂ O ₂)	Sigma Aldrich	Cat#: H1009
HEPES	Life Technologies	Cat# 15630056
Collagenase IV	Sigma Aldrich	Cat#: C5138
HBSS	Sigma Aldrich	Cat#: H9394
RPMI 1640	Sigma Aldrich	Cat#: R0883
Fetal calf serum (FCS)	Biochrom	N/A
Dimethyl sulfoxide (DMSO)	Sigma Aldrich	Cat#: D8418
Propidium Iodide (PI)	Thermo Fisher	Cat#: P1304MP

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Critical commercial assays		
High Sensitivity DNA Kit	Agilent	Cat#: 5067-4626
Qubit dsDNA hs assay kit	Life Technologies	Cat#: Q32854
Chromium Next GEM Single Cell 5' Kit v2	10x Genomics	Cat#: 1000263
ZellSafe Chip kit FFPE	Canopy Biosciences	Cat#: 28050606/04
Deposited data		
Healthy human control dataset	James et al. ³⁹	https://www.gutcellatlas.org
GvHD scRNA seq dataset	This publication	GEO: GSE234357
Microbiome 16S rRNA dataset	This publication	ENA: PRJEB60178
Software and algorithms		
ImageJ 1.53c	Schindelin et al. ⁶⁶	https://imagej.net/software/fiji/
Affinity photo V1.8.3	Serif Europe Ltd. 2020	https://affinity.serif.com/
FlowJo 10	FlowJo LLC	https://www.flowjo.com
Prism 9	Graphpad	https://www.graphpad.com
Scanpy 1.8	Wolf et al. ⁶⁷	https://github.com/theislab/scanpy
Matlab R2018b	Mathworks	https://de.mathworks.com/products/matlab.html
Automated fluorescence signal quantification	Jarosch et al. ³⁵	https://github.com/SebastianJarosch/ChipCytometry-Image-Processing
Cell Ranger V6.0.2	10x Genomics	https://support.10xgenomics.com/single-cell-gene-expression/software/pipelines/latest/installation
Souporcell V2.0	Heaton et al. ⁶⁸	https://github.com/wheaton5/souporcell
arcasHLA V0.4	Orenbuch et al. ⁶⁹	https://github.com/RabadanLab/arcasHLA
Diffxpy	David Fischer & Florian Hölzlwimmer	https://github.com/theislab/diffxpy
Scirpy V0.10.0	Sturm et al. ⁷⁰	https://github.com/scverse/scirpy
Velocity V0.17	La Manno et al. ⁷¹	https://github.com/velocity-team/velocity.py
scVelo V0.2.3	Bergen et al. ⁷²	https://github.com/theislab/scvelo
Scipy V1.6.1	Virtanen et al. ⁷³	https://scipy.org/
Seaborn V0.10.0	Waskom ⁷⁴	https://seaborn.pydata.org/
Matplotlib V3.1.3	Hunter ⁷⁵	https://matplotlib.org/
TorrentSuite V5.16.1	Thermo Fisher Scientific	-
Trimmomatic V0.9	Bolger et al. ⁷⁶	https://github.com/usadellab/Trimmomatic
Cutadapt V3.5	Marcel Martin	https://github.com/marcelm/cutadapt/
Scikit-posthocs V0.7	Terpilowski ⁷⁷	https://github.com/maximtrp/scikit-posthocs
dada2 V1.16	Callahan et al. ⁷⁸	https://github.com/benjjneb/dada2

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Dirk H. Busch (dirk.busch@tum.de).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- Single-cell RNA-seq and microbiome 16S rRNA sequencing data have been deposited at GEO and European Nucleotide Archive (ENA), respectively, and are publicly available as of the date of publication. Accession numbers are listed in the [key resources table](#).
- This paper does not report original code.

- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Description of the study cohort

GI biopsies were sampled from different gut anatomical locations of patients receiving aHSCT at different time points after the transplant or/and the development of GvHD, and assessed by experienced pathologists blinded to the clinical data. GI-aGvHD was graded according to the Lerner grading system⁷⁹ whereas clinical GvHD at the time of biopsy was graded according to the Glucksberg grading system³⁷. For data analyses, biopsies were grouped according to GvHD severity into mild (clinical GvHD score in the gastrointestinal tract 0-1) and severe (clinical GvHD score in the gastrointestinal tract 2-4). For additional details, see [Table S1](#). All procedures were approved by the local ethics committee (ethical committee of the University of Regensburg - 09/059 and 18-684482-101). All patients gave informed consent and the studies were performed under the regulation of Helsinki. The cohort size was defined according to the maximum availability of samples.

METHOD DETAILS

Clinical data

Steroid treatment at the time of biopsy was defined as treatment with systemic steroids (≥ 7.5 mg/d prednisolone) at least 24 h before the biopsy was obtained. Steroid refractory GvHD was defined as previously described⁸⁰ (either no response to steroid therapy or inability to taper steroids without GvHD worsening). Antibiotic exposure (ABX, Piperacillin/Tazobactam +/- Vancomycin as first line therapy, or Meropenem +/- Vancomycin as second line treatment) at the time of biopsy was defined as systemic exposure with broad-spectrum antibiotics at least 24 h ahead of biopsy sampling. The biopsy was assigned "no ABX" when no broad-spectrum antibiotics were given at the time of biopsy or discontinued a minimum of 14 days before the biopsy. Serum biomarkers ST2, Reg3a, and MAP were measured via enzyme-linked immunosorbent assays (ELISA) as previously described⁸¹ at the onset of GvHD symptoms and, where available, also at the time of biopsy. Mean time between available onset and biopsy samples was 11 days (0-70). Stool samples were taken at least in the week of biopsy, for most of the samples on the same day.

Generation of single-cell suspension from human GI biopsies

Biopsies were digested for 30 min at 37°C using 1.5 ml digestion buffer (25 mM HEPES, 0.1 % Collagenase IV in HBSS w/o Ca^{2+} , Mg^{2+}) immediately after resection. After 15 min of digestion, tissues were vortexed again to maximize mechanical dissociation. The suspension was filtered through a 100 μm nylon mesh, and digestion was stopped by adding 25 ml wash buffer (RPMI 1640, 10 % FCS, 25 mM HEPES). Centrifugation (470 x g, RT) was followed by another washing step with 25 ml wash buffer before the cells were resuspended in an appropriate volume of RPMI medium (10 % FCS) for cell counting.

Cells were frozen in 250 μl freezing media (FCS, 10% DMSO) and cooled down to -80°C at a speed of 1°C/min before they were transferred to liquid nitrogen for long-term storage. At the day of sorting, cells were thawed in 10 ml pre-warmed (37°C) RPMI medium by transferring the warm media directly to the frozen cells. Cells were washed in 10 ml FACS buffer (PBS, 0.5 % BSA) once before proceeding with antibody staining.

Antibody staining and flow cytometry-based cell sorting

Cells from GI biopsies were stained with anti-CD45 antibody in 100 μl FACS buffer for 30 min on ice in a master mix also containing the respective hashtag antibody and the DNA-barcoded surface antibody cocktail according to the manufacturer's recommendation. Staining was followed by two washing steps with 200 μl FACS buffer each before living, CD45 positive cells were enriched via fluorescence-activated cell sorting (FACS). Live-dead staining (propidium iodide) was added to the cells just before the sort to minimize the number of dead cells in the downstream process. Several samples stained with different hashtag antibodies (always from different donors) were pooled together by sorting them into the same well of a 96-well plate. A maximum of 6.000 cells was sorted per biopsy.

5' single-cell RNA sequencing of human GI biopsies

After sorting, living CD45⁺ cells were centrifuged, and the supernatant was carefully removed. Cells were resuspended in the 37.2 μl master mix + 37.8 μl water before 70 μl of the cell suspension were transferred to the chip. After each step, the integrity of the pellet was checked under the microscope to ensure that all cells were loaded onto the chip. From here on, 10x experiments have been performed according to the manufacturer's protocol. Quality control has been performed with a High Sensitivity DNA Kit on a Bioanalyzer 2100 as recommended in the protocol, and libraries were quantified with the Qubit dsDNA hs assay kit. All steps have been performed using RPT filter tips and DNA LoBind tubes. For sequencing, libraries have been pooled according to their minimal required read counts (20.000 reads/cell for gene expression libraries and 5.000 reads/cell for TCR/surface antibody libraries). Illumina paired-end sequencing was performed with 28+91 bp on a HiSeq2500 or with 2x150 bp on a NovaSeq 6000. Annotation against the human genome (GRCh38) and a corresponding VDJ reference was performed using Cell Ranger (10x genomics).

ChipCytometry of FFPE tissues

ChipCytometry of human FFPE biopsies was performed according to our in-house developed procedure.^{35,36} Tissue sections (4–5 μ m) were mounted on glass coverslips (24 x 50 mm, 1 mm thickness, Engelbrecht Automat-Star #K12460A1,0) and dried for at least 10 h. Paraffin was melted by overnight incubation at 60°C, followed by heating at 70°C for additionally 30 min. Tissue slides were then de-paraffinized and rehydrated via the following incubations: 3 x 10 min Xylene, 2 x 10 min absolute ethanol, 1 x 5 min ethanol 90%, 1 x 5 min ethanol 70%, 1 x 5 min ethanol 50%, 1 x 5 min tap water. The antigen retrieval was performed in a water bath containing a dish with a basic retrieval buffer (10 mM TRIS, 1 mM EDTA, pH 8.5) at 90°C for 20 min. The coverslips were cooled down in PBS at RT, loaded onto ZellSafe Chips (ZELLKRAFTWERK) according to manufacturer's recommendations, and incubated with 500 μ l goat serum (5% in PBST) for 1 h at RT for blocking. For decreasing autofluorescence, chips were rinsed with 500 μ l Sudan Black B solution (0.1% in 70% EtOH, Sigma Aldrich, #199664), incubated for 10 min and extensively washed with 70% EtOH and PBST (0.1% Tween-20 in PBS). A first background was acquired in the FITC channel to locate the tissue on the chip and select the desired positions. A background was acquired for each channel before each cycle of staining.

For staining, mixes of fluorophore-conjugated primary antibodies were centrifuged for 10 min at 14,000 rpm, and 300 μ l of the antibody cocktail were incubated overnight at 4°C, washed and acquired in the respective channel. Nuclei staining was performed via Hoechst (Thermo Scientific, H3570), diluted 1:50,000 in PBST for 5 min at RT. Secondary staining of rabbit anti-CD3 was performed for 2 h at RT. Images were acquired according to the following exposure times: 300 ms for PE, 500 ms for FITC, 300 ms for PerCP-Cy5.5, 50 ms for BV421 and 1000 ms for BUV395. After imaging, photobleaching was performed with the ChipCytometry microscope by exposing each position of the tissue for 20 seconds to white light from the build-in HBO lamp with a 364 nm longpass filter to protect epitope damage from UV light.

16S rRNA sequencing

Nucleic acids were extracted from 50 mg patient stool collected and stabilized in maGix microbiome preservation buffer (microBIO-Mix). Lysis of microbial cells was enhanced by repeated beat-beating of stool suspensions on a Tissue Lyser II instrument (Qiagen) using Lysing Matrix Y beads (MP Biomedicals) followed by freezing and thawing. Total nucleic acids were purified from crude lysates with the MagNA Pure 96 system (Roche). Bacterial 16S rRNA gene copies were quantified by qPCR.⁸² V1-3 hypervariable regions of bacterial 16S rRNA genes were amplified from 1×10^7 16S copies with the Platinum II Taq Hot-Start DNA-Polymerase and primers S-D-Bact-0008-c-S-20 and S-D-Bact-0517-a-A-18.⁸³ Amplicon libraries were purified with MagSi-NGSPREP Plus beads (Magtivity), quantified with the KAPA Library Quantification Kit (Roche) before normalization and pooling. Final library pools were sequenced on a GeneStudio S5 Plus instrument (Thermo Fisher Scientific) with the 600 bp protocol after isothermal amplification on an IonChef instrument.

QUANTIFICATION AND STATISTICAL ANALYSIS

Data processing of single-cell RNA sequencing data

Annotation against the human reference genome (GRCh38/Ensembl 98) and the related VDJ reference was performed using Cell Ranger. All subsequent analyses have been performed using SCANPY.⁶⁷ The control dataset from the human gut cell atlas³⁹ was imported from the Cell Ranger output files, down-sampled to a maximum of 1000 cells per sample to fit the biopsy sizes and included in all pre-processing steps in parallel to our samples. Cells with less than 200 genes and with more than 20% mitochondrial gene expression were excluded from the analysis as well as genes present in less than three cells. Biopsies with less than 30 cells were excluded from the analysis. General data pre-processing was performed according to good practice in scRNA seq analysis⁸⁴ with the following modifications: since the dataset contained plasma cells that are bigger in size compared to the other cell types, filtering was performed according to the factor between the number of counts and the number of genes per cells. The number of counts, the fraction of mitochondrial genes, and cell cycle scores were regressed out, and batch effect correction was performed using batch-balanced k-nearest neighbors (bbknn). Surface antibody reads were normalized by centered log-ratio transformation and scaled for each cell. Cell type annotation was based on Leiden clustering combined with sub-clustering of heterogeneous clusters and final annotation with known marker genes.

Demultiplexing of single-cell RNA sequencing data

Donor demultiplexing was performed using souporecell,⁶⁸ followed by HLA genotype prediction for each cluster by arcasHLA.⁶⁹ An HLA matching score taking HLA matching, gender and cell numbers into account was utilized to finally map each donor to a cluster. For barcoded libraries, hashsolo⁸⁵ was used for demultiplexing.

Gene set scoring

Previously defined genes associated with regulatory or effector T cells⁴³ or genes associated with Treg functionality defined according to literature research (FOXP3,⁸⁶ IL2RA,⁸⁷ CTLA4,⁸⁸ CD27,⁸⁹ TNFRSF4, TNFRSF18, TNFRSF9,⁹⁰ TIGIT,⁹¹ BATF,⁹² CCR4,⁹³ CXCR3⁹⁴) have been scored by calculating the average expression of the set of genes subtracted with the average expression of a set of 50 randomly sampled reference genes. The genes for the individual scores can be found in Figure S4 (only overlapping genes between the published set and our dataset are listed).

Definition of cell types for individual analysis

Tregs (Figure 3) were defined as FOXP3 and CD3D double positive cells for analysis. Conventional T cells (Figure 5) were defined as cells from annotated T cell clusters for which also a T cell receptor was detected. All cells fitting these criteria were used for in-depth analysis of the individual compartments.

Differential gene expression and gene set enrichment analysis

Differential gene expression analysis was performed using Welch's t-test in the `diffxpy` python package. Gene set enrichment analysis was performed with `GSEAPy`. Gene set reference databases are depicted in the figure and the top 10 terms with a p value below 0.05 are depicted.

Clonotype definition from single-cell RNA sequencing data

Clonotype analysis was performed using `scirpy`.⁷⁰ Cells belonging to one clonotype were defined to have identical TRA- and TRB-CDR3 nucleotide sequences and both pairs of TRA/TRB sequences were considered in case additional chains were present.

TCR similarity analysis

TCR similarity analysis was performed with `GLIPH` or `TCR dist`, both based on CDR3 amino acid sequences. `GLIPH` analysis was performed on the TCR beta sequence only, while `TCR dist` measurements were performed with `scirpy` on paired α/β sequences.

RNA velocity analysis

RNA velocities were calculated from Cell ranger output data using `velocyto`⁷¹ and the results were analyzed with `scVelo`⁷² using a dynamic model of velocity estimation.

ChipCytometry data analysis

ChipCytometry data were analyzed in an automated manner as described before.^{35,36} In brief, nuclei segmentation was followed by spatial spillover correction and calculation of the mean intensity for each cell and each marker. Cells were then classified according to their marker expression via population gating, and infiltration densities were defined as the detected number of cells from a defined subset divided by the size of the biopsy.

Analysis of variance determining factors

Principal component analysis (PCA) was performed using `sklearn` in Python on the normalized set of features present in most of the biopsies (30 biopsies, features: T cell/mm², CD4/mm², CD8/mm², Tregs/mm², Leukocytes/mm², activated Tregs (% of Tregs), CD4/CD8 ratio). Analysis of variance-determining factors was performed via calculating the foldchange on a principal component axis regarding the value of a clinical parameter. Non-binary parameters were binned according to the mean of the clinical parameter across the dataset into low and high values. Values for the individual clinical parameters can be found in Table S4.

Replotting of imaging data

Replotting (Figures 1D and 6C) was achieved by plotting all cells according to their x and y coordinates and color coding each cell according to its cell type. The color code resulted from plotting the individual gated populations (colored) on top of the ungated cells in the background (gray).

Spatial distance analysis

Distances were calculated according to the coordinates of the center point for each cell. In Figure 6D, for each CD8 T cell, the distance to all regulatory T cells was calculated. The minimum distance resulted in a distance to the closest Treg for each CD8 T cell.

16S rRNA microbiome data

Raw sequencing reads were retrieved from `TorrentSuite` (5.16.1) followed by quality trimming with `Trimmomatic` (0.9). Sequencing adapters and 16S primer sequences were removed and demultiplexed using `cutadapt` (3.5). Amplicon sequence variants (ASV) were retrieved from processed reads in R (4.1.1) using the `dada2` (1.16) package. Here, 15 bp were trimmed from 5'-ends of each read, and the `dada2` command was executed while a `minEE` value of 5, a `HOMOPOLYMER_GAP_PENALTY` of -1 and a `BAND_SIZE` of 32 were applied. Taxonomy was assigned to resulting ASVs with the `IDTAXA` classifier of the `DECIPHER` package (2.19) and the `All-Species Living Tree (LTP)` database (12.2021) applying a 50% confidence threshold. Correlation analysis was performed based on the abundance of assigned taxonomic features.

Association of taxonomic features with Treg phenotypes

Patients were grouped according to the presence of a taxonomic feature for each taxon and the foldchange of the median score was calculated between the patient groups for the Treg gene score, the `Teff` gene score and the Treg suppression score. Bacteria were then clustered via K-means clustering according to their influence on the three scores.

Visualization and statistical testing

Data were displayed using GraphPad Prism (V9.1) or the seaborn (0.10.0) and matplotlib (3.1.3) packages in python. Boxplots always depicted the median of the data and interquartile range (IQR), with whiskers extending to 1.5 times IQR. Statistical and correlation analyses were performed with the stats.linregress method from the scipy (1.4.1) module, calculating linear least-squares regressions for two sets of measurements. For linear regressions, p-values were calculated for a hypothesis test whose null hypothesis was that the slope is zero, using Wald Test with t-distribution of the test statistic. Pearson correlation coefficients were depicted as r-values. Significance was defined as n.s. = not significant, *p-value < 0.05, **p-value < 0.01, ***p-value < 0.001, ****p-value < 0.0001. Information on statistical tests used for individual figures can be found in the figure legends. All samples (31 fresh biopsies for scRNA-seq and 51 FFPE biopsies for ChipCytometry) have been included in the analyses unless stated otherwise in the figure legend.